

Yu-Hsiang Wang

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Education

University of Illinois Urbana-Champaign

PH.D. IN ELECTRICAL & COMPUTER ENGINEERING

Champaign, IL

Aug. 2024 - Present

- **Advisor :** Olga Milenkovic
- **Overall GPA :** 4.00/4.00

National Taiwan University

B.S. IN ELECTRICAL ENGINEERING

Taipei, Taiwan

Sep. 2019 - Jan. 2024

- **Overall GPA :** 4.09/4.30

Research Experience

University of Illinois Urbana-Champaign

GRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. OLGICA MILENKOVIC

Champaign, IL

Aug. 2024 - Present

- **Time Series Generation:** Proposed WaveletDiff, a multiresolution diffusion model for time series data generation via Discrete Wavelet Transforms. [TMLR 2026]
- **Transition Path Sampling:** Designed stochastic control policies utilizing probabilistic bias fields to overcome initialization sensitivity in robust molecular transition path sampling. [arXiv 2026]
- **Molecule Generation:** Proposed DMol, a schedule-driven graph diffusion model for highly efficient and versatile molecule generation. [NeurIPS 2025]
- **Nanopore Signal Processing:** Developed a machine-learning framework utilizing wavelet feature extraction to accurately classify Abl kinase inhibitor binding types from single-molecule nanopore tweezers.

University of California, Los Angeles

RESEARCH ASSISTANT, ADVISOR: PROF. CHO-JUI HSIEH

Los Angeles, CA

Jun. 2023 - Sep. 2023

- **LLM Uncertainty:** Proposed CLUE, a novel concept-level uncertainty estimation framework for Large Language Models to improve reliability and output evaluation. [NeurIPS 2024 SFLLM workshop]

Publication

- [1] Ly Nguyen, **Yu-Hsiang Wang**, Joshua Foster, David DeCoeur, Lan Nguyen, Bryant Wu, Olga Milenkovic, and Min Chen. "Subtyping Abl Kinase Inhibitor Binding Modes with Machine-Learning-Enabled Super Resolution Single-Molecule Nanopore Tweezers". *In Submission, bioRxiv 2026*.
- [2] Jingqian Liu, **Yu-Hsiang Wang**, Yanru Qu, and Ge Liu. "Stochastic Control Policies for Robust Molecular Transition Path Sampling". *In Submission, arXiv 2026*.
- [3] **Yu-Hsiang Wang** and Olga Milenkovic. "WaveletDiff: Multilevel Wavelet Diffusion For Time Series Generation". *TMLR 2026*.
- [4] Peizhi Niu, **Yu-Hsiang Wang**, Vishal Rana, Chetan Rupakheti, Abhishek Pandey, and Olga Milenkovic. "DMol: A Highly Efficient and Chemical Motif-Preserving Molecule Generation Platform". *NeurIPS 2025*.
- [5] Ming-Yi Hong, Miao-Chen Chiang, Youchen Teng, **Yu-Hsiang Wang**, Chih-Yu Wang, and Che Lin. "TheGAU: Type-Aware Heterogeneous Graph Autoencoder and Augmentation". *In Submission, arXiv 2025*.
- [6] **Yu-Hsiang Wang**, Andrew Bai, Che-Ping Tsai, and Cho-Jui Hsieh. "CLUE: Concept-Level Uncertainty Estimation for Large Language Models". *NeurIPS 2024 SFLLM Workshop*.

Honors & Awards

Best Presentation Award, CSL Student Conference (CSLSC)

Feb 2026

Research Grant for College Students, National Science and Technology Council (Taiwan)

July 2023 - Feb 2024

Dean's List Award, Electrical Engineering Dept. at NTU

Fall 2020

Teaching

ECE313: Probability with Engineering Applications

GRADUATE TEACHING ASSISTANT

UIUC

Fall 2025